

1. Abstract

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

2,633 nodules; 868 patients; folds 479/ 502/ 539/ 549/ 564; patient leakage 0.

Grad-CAM: 73,724 requested = 66,769 valid + 6,955 undefined post-ReLU zero maps.

Bootstrap: 2,000 patient-cluster draws.

Black-box: MAE 0.5006 [0.4829, 0.5195]; RMSE 0.6422; AUROC 0.9453; AUPRC 0.8937.

Standard CBM: MAE 0.5021 [0.4828, 0.5223]; RMSE 0.6496; AUROC 0.9327; AUPRC 0.8657.

Mixed-type CEM: MAE 0.4841 [0.4669, 0.5021]; RMSE 0.6283; AUROC 0.9417; AUPRC 0.8774.

Learned-softmax GAM: MAE 0.4804 [0.4623, 0.4985]; RMSE 0.6176; AUROC 0.9493; AUPRC 0.9026.

2. Introduction

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

LIDC malignancy is a radiologist assessment rather than pathology-confirmed diagnosis [1].

The system is a research benchmark and is not a clinical diagnostic product.

The models used DenseNet-121 [2]; CBM and CEM terminology follows [3], [4], while Mixed-type CEM and Learned-softmax GAM denote the preregistered project-specific designs.

3. Methods

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Primary scores were used without clipping; 2,000 shared patient-cluster bootstrap draws produced percentile 95% CIs [6].

Secondary Youden-J thresholds used only fold-specific validation samples with malignancy ≤ 2 or ≥ 4 .

Each valid Grad-CAM map [5] used 26,215 saliency voxels and 20 matched random masks.

Paired signs are $\Delta\text{MAE} = \text{MAE}_A - \text{MAE}_B$ and $\Delta\text{AUROC} = \text{AUROC}_B - \text{AUROC}_A$; positive values favor model B.

Intervention signs are $\Delta_i\text{MAE} = \text{baseline_MAE} - i\text{MAE}$ and $\Delta_i\text{AUC} = i\text{AUC} - \text{baseline_AUROC}$; positive values denote improvement.

4. Cohort and Integrity

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The canonical OOF cohort contained 2,633 unique nodules from 868 patients.

Outer test fold counts were 479/ 502/ 539/ 549/ 564; patient leakage was 0.

All 4 models used identical targets and fold-specific outer test membership.

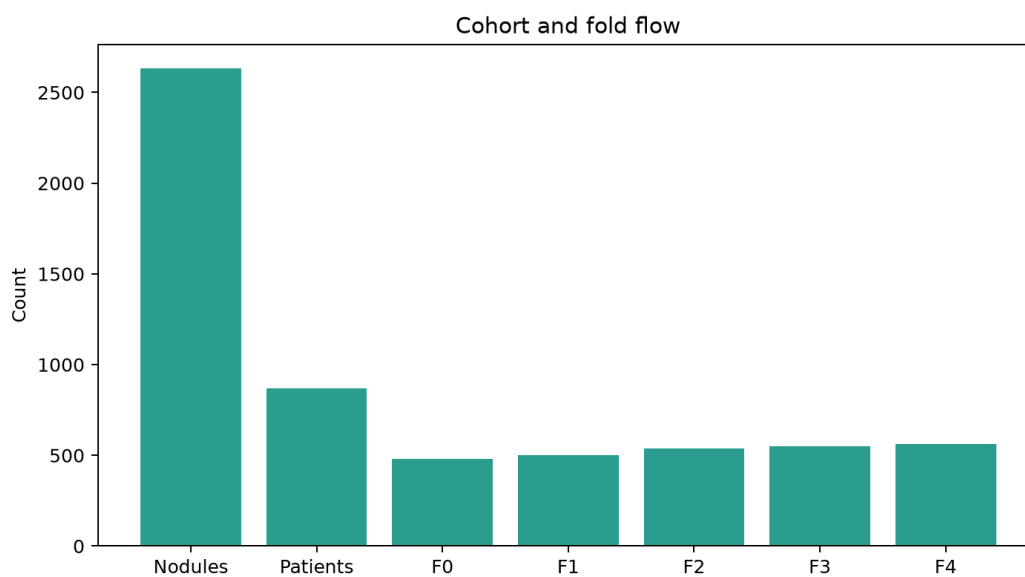


Figure 1

5. Primary and Secondary Results

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Black-box: MAE 0.5006 [95% CI 0.4829, 0.5195]; RMSE 0.6422 [95% CI 0.6189, 0.6671]; normalized MAE 0.1252 [95% CI 0.1207, 0.1299]; Pearson 0.7157 [95% CI 0.6894, 0.7409]; Spearman 0.6345 [95% CI 0.5994, 0.6676]; unclipped 1-5 prediction range [0.4885, 5.1200]; below 1 rate 0.0148; above 5 rate 0.0011.

Standard CBM: MAE 0.5021 [95% CI 0.4828, 0.5223]; RMSE 0.6496 [95% CI 0.6254, 0.6749]; normalized MAE 0.1255 [95% CI 0.1207, 0.1306]; Pearson 0.7076 [95% CI 0.6770, 0.7354]; Spearman 0.6091 [95% CI 0.5700, 0.6482]; unclipped 1-5 prediction range [0.8585, 4.5798]; below 1 rate 0.0087; above 5 rate 0.0000.

Mixed-type CEM: MAE 0.4841 [95% CI 0.4669, 0.5021]; RMSE 0.6283 [95% CI 0.6040, 0.6537]; normalized MAE 0.1210 [95% CI 0.1167, 0.1255]; Pearson 0.7296 [95% CI 0.7010, 0.7565]; Spearman 0.6400 [95% CI 0.6036, 0.6734]; unclipped 1-5 prediction range [0.8229, 4.9351]; below 1 rate 0.0042; above 5 rate 0.0000.

Learned-softmax GAM: MAE 0.4804 [95% CI 0.4623, 0.4985]; RMSE 0.6176 [95% CI 0.5924, 0.6423]; normalized MAE 0.1201 [95% CI 0.1156, 0.1246]; Pearson 0.7405 [95% CI 0.7117, 0.7676]; Spearman 0.6528 [95% CI 0.6160, 0.6883]; unclipped 1-5 prediction range [1.0081, 4.6816]; below 1 rate 0.0000; above 5 rate 0.0000.

DeltaMAE Black-box - Learned-softmax GAM = 0.0201 [95% CI 0.0101, 0.0305]; crosses zero: no.

DeltaMAE Black-box - Mixed-type CEM = 0.0164 [95% CI 0.0060, 0.0272]; crosses zero: no.

DeltaMAE Black-box - Standard CBM = -0.0016 [95% CI -0.0148, 0.0120]; crosses zero: yes.

DeltaMAE Mixed-type CEM - Learned-softmax GAM = 0.0037 [95% CI -0.0059, 0.0130]; crosses zero: yes.

DeltaMAE Standard CBM - Learned-softmax GAM = 0.0217 [95% CI 0.0109, 0.0326]; crosses zero: no.

DeltaMAE Standard CBM - Mixed-type CEM = 0.0181 [95% CI 0.0059, 0.0301]; crosses zero: no.

Black-box: AUROC 0.9453 [95% CI 0.9264, 0.9615]; AUPRC 0.8937 [95% CI 0.8586, 0.9250].

Standard CBM: AUROC 0.9327 [95% CI 0.9113, 0.9510]; AUPRC 0.8657 [95% CI 0.8263, 0.8995].

Mixed-type CEM: AUROC 0.9417 [95% CI 0.9202, 0.9605]; AUPRC 0.8774 [95% CI 0.8332, 0.9155].

Learned-softmax GAM: AUROC 0.9493 [95% CI 0.9266, 0.9685]; AUPRC 0.9026 [95% CI 0.8678, 0.9340].

DeltaAUROC Learned-softmax GAM - Black-box = 0.0042 [95% CI -0.0051, 0.0135]; crosses zero: yes.

DeltaAUROC Mixed-type CEM - Black-box = -0.0036 [95% CI -0.0128, 0.0052]; crosses zero: yes.

DeltaAUROC Standard CBM - Black-box = -0.0127 [95% CI -0.0219, -0.0036]; crosses zero: no.

DeltaAUROC Learned-softmax GAM - Mixed-type CEM = 0.0078 [95% CI -0.0008, 0.0178]; crosses zero: yes.

DeltaAUROC Learned-softmax GAM - Standard CBM = 0.0169 [95% CI 0.0064, 0.0273]; crosses zero: no.

DeltaAUROC Mixed-type CEM - Standard CBM = 0.0091 [95% CI -0.0020, 0.0203]; crosses zero: yes.

6. Concept Prediction Results

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

Standard CBM / subtlety: MAE 0.1690; RMSE 0.2233; Pearson 0.5673; Spearman 0.5733; N 2633.
Standard CBM / internalStructure: soft CE 0.0390; Brier 0.0069; macro-F1 0.2497; soft N 2633; hard N 2625; ties 8.
Standard CBM / calcification: soft CE 0.2072; Brier 0.0491; macro-F1 0.3138; soft N 2633; hard N 2578; ties 55.
Standard CBM / sphericity: MAE 0.1386; RMSE 0.1767; Pearson 0.4366; Spearman 0.4528; N 2633.
Standard CBM / margin: MAE 0.1456; RMSE 0.1972; Pearson 0.6868; Spearman 0.6473; N 2633.
Standard CBM / lobulation: MAE 0.1310; RMSE 0.1857; Pearson 0.4597; Spearman 0.4559; N 2633.
Standard CBM / spiculation: MAE 0.1211; RMSE 0.1816; Pearson 0.5069; Spearman 0.4164; N 2633.
Standard CBM / texture: MAE 0.1161; RMSE 0.1802; Pearson 0.7828; Spearman 0.5806; N 2633.
Mixed-type CEM / subtlety: MAE 0.2038; RMSE 0.2502; Pearson 0.4445; Spearman 0.4491; N 2633.
Mixed-type CEM / internalStructure: soft CE 0.0827; Brier 0.0136; macro-F1 0.2497; soft N 2633; hard N 2625; ties 8.
Mixed-type CEM / calcification: soft CE 0.2624; Brier 0.0677; macro-F1 0.3098; soft N 2633; hard N 2578; ties 55.
Mixed-type CEM / sphericity: MAE 0.1780; RMSE 0.2171; Pearson 0.0561; Spearman 0.0514; N 2633.
Mixed-type CEM / margin: MAE 0.2463; RMSE 0.2921; Pearson 0.1880; Spearman 0.1709; N 2633.
Mixed-type CEM / lobulation: MAE 0.1829; RMSE 0.2175; Pearson 0.2701; Spearman 0.2488; N 2633.
Mixed-type CEM / spiculation: MAE 0.1799; RMSE 0.2155; Pearson 0.3050; Spearman 0.2743; N 2633.
Mixed-type CEM / texture: MAE 0.2650; RMSE 0.3052; Pearson 0.2025; Spearman 0.1666; N 2633.
Learned-softmax GAM / subtlety: MAE 0.1677; RMSE 0.2209; Pearson 0.5785; Spearman 0.5823; N 2633.
Learned-softmax GAM / internalStructure: soft CE 0.0381; Brier 0.0073; macro-F1 0.3121; soft N 2633; hard N 2625; ties 8.
Learned-softmax GAM / calcification: soft CE 0.2013; Brier 0.0481; macro-F1 0.3127; soft N 2633; hard N 2578; ties 55.
Learned-softmax GAM / sphericity: MAE 0.1400; RMSE 0.1784; Pearson 0.4242; Spearman 0.4351; N 2633.
Learned-softmax GAM / margin: MAE 0.1455; RMSE 0.1983; Pearson 0.6834; Spearman 0.6478; N 2633.
Learned-softmax GAM / lobulation: MAE 0.1272; RMSE 0.1896; Pearson 0.4509; Spearman 0.4721; N 2633.
Learned-softmax GAM / spiculation: MAE 0.1144; RMSE 0.1838; Pearson 0.4985; Spearman 0.4561; N 2633.
Learned-softmax GAM / texture: MAE 0.1112; RMSE 0.1791; Pearson 0.7861; Spearman 0.5829; N 2633.

7. Concept Intervention

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

Standard CBM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.5021/ 0.5010/ 0.5001/ 0.4998/ 0.4996/ 0.4998/ 0.5023/ 0.5041/ 0.5075.

Standard CBM permutation iMAE 0.5014; Delta_iMAE 0.0006 (positive denotes improvement).

Standard CBM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9327/ 0.9321/ 0.9311/ 0.9304/ 0.9295/ 0.9287/ 0.9271/ 0.9257/ 0.9237.

Standard CBM permutation iAUC 0.9291; Delta_iAUC -0.0036 (positive denotes improvement).

Standard CBM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.5021/ 0.4952/ 0.5082/ 0.5070/ 0.5096/ 0.5080/ 0.5075/ 0.5074/ 0.5075.

Standard CBM iMAE 0.5060; Delta_iMAE -0.0039 (positive denotes improvement).

Standard CBM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9327/ 0.9265/ 0.9186/ 0.9166/ 0.9181/ 0.9215/ 0.9236/ 0.9239/ 0.9237.

Standard CBM iAUC 0.9221; Delta_iAUC -0.0106 (positive denotes improvement).

Mixed-type CEM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.4841/ 0.4755/ 0.4681/ 0.4601/ 0.4540/ 0.4483/ 0.4434/ 0.4391/ 0.4358.

Mixed-type CEM permutation iMAE 0.4560; Delta_iMAE 0.0281 (positive denotes improvement).

Mixed-type CEM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9417/ 0.9456/ 0.9493/ 0.9523/ 0.9549/ 0.9573/ 0.9598/ 0.9617/ 0.9638.

Mixed-type CEM permutation iAUC 0.9542; Delta_iAUC 0.0125 (positive denotes improvement).

Mixed-type CEM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.4841/ 0.4585/ 0.4484/ 0.4425/ 0.4400/ 0.4367/ 0.4352/ 0.4356/ 0.4358.

Mixed-type CEM iMAE 0.4446; Delta_iMAE 0.0395 (positive denotes improvement).

Mixed-type CEM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9417/ 0.9529/ 0.9593/ 0.9600/ 0.9618/ 0.9631/ 0.9637/ 0.9637/ 0.9638.

Mixed-type CEM iAUC 0.9597; Delta_iAUC 0.0180 (positive denotes improvement).

Learned-softmax GAM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.4804/ 0.4775/ 0.4764/ 0.4760/ 0.4787/ 0.4827/ 0.4891/ 0.4956/ 0.5068.

Learned-softmax GAM permutation iMAE 0.4837; Delta_iMAE -0.0033 (positive denotes improvement).

Learned-softmax GAM 100-permutation mean k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9493/ 0.9493/ 0.9491/ 0.9479/ 0.9450/ 0.9423/ 0.9388/ 0.9354/ 0.9298.

Learned-softmax GAM permutation iAUC 0.9434; Delta_iAUC -0.0059 (positive denotes improvement).

Learned-softmax GAM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 MAE: 0.4804/ 0.4645/ 0.4901/ 0.4998/ 0.5049/ 0.5059/ 0.5070/ 0.5068/ 0.5068.

Learned-softmax GAM iMAE 0.4966; Delta_iMAE -0.0161 (positive denotes improvement).

Learned-softmax GAM error-first k=0/ 1/ 2/ 3/ 4/ 5/ 6/ 7/ 8 AUROC: 0.9493/ 0.9464/ 0.9322/ 0.9258/ 0.9245/ 0.9282/ 0.9298/ 0.9297/ 0.9298.

Learned-softmax GAM iAUC 0.9320; Delta_iAUC -0.0173 (positive denotes improvement).

8. Grad-CAM Accounting

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

Requested 73724 = valid 66769 + undefined 6955; undefined rate 0.0943.

Black-box: valid 2429; undefined 204; rate 0.0775.

Black-box fold 0: valid 479; undefined 0; rate 0.0000.

Black-box fold 1: valid 502; undefined 0; rate 0.0000.

Black-box fold 2: valid 539; undefined 0; rate 0.0000.

Black-box fold 3: valid 448; undefined 101; rate 0.1840.

Black-box fold 4: valid 461; undefined 103; rate 0.1826.

Standard CBM: valid 22413; undefined 1284; rate 0.0542.

Standard CBM fold 0: valid 3817; undefined 494; rate 0.1146.

Standard CBM fold 1: valid 4218; undefined 300; rate 0.0664.

Standard CBM fold 2: valid 4591; undefined 260; rate 0.0536.

Standard CBM fold 3: valid 4888; undefined 53; rate 0.0107.

Standard CBM fold 4: valid 4899; undefined 177; rate 0.0349.

Mixed-type CEM: valid 20316; undefined 3381; rate 0.1427.

Mixed-type CEM fold 0: valid 3474; undefined 837; rate 0.1942.

Mixed-type CEM fold 1: valid 3992; undefined 526; rate 0.1164.

Mixed-type CEM fold 2: valid 4182; undefined 669; rate 0.1379.

Mixed-type CEM fold 3: valid 4735; undefined 206; rate 0.0417.

Mixed-type CEM fold 4: valid 3933; undefined 1143; rate 0.2252.

Learned-softmax GAM: valid 21611; undefined 2086; rate 0.0880.

Learned-softmax GAM fold 0: valid 4067; undefined 244; rate 0.0566.

Learned-softmax GAM fold 1: valid 4298; undefined 220; rate 0.0487.

Learned-softmax GAM fold 2: valid 4515; undefined 336; rate 0.0693.

Learned-softmax GAM fold 3: valid 4064; undefined 877; rate 0.1775.

Learned-softmax GAM fold 4: valid 4667; undefined 409; rate 0.0806.

The complete model x fold x target/ concept breakdown is retained in Table 7 (gradcam_accounting.csv).

9. Spatial Faithfulness

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

Black-box output_sensitivity: saliency mean 0.0252; saliency-random mean -0.3225; 95% range [-0.7498, -0.0148]; saliency > random mean rate 0.0086.

Black-box error_increase: saliency mean 0.0029; saliency-random mean -0.2354; 95% range [-0.7076, 0.1958]; saliency > random mean rate 0.1441.

Standard CBM output_sensitivity: saliency mean 0.1494; saliency-random mean -1.0642; 95% range [-3.2466, 0.0454]; saliency > random mean rate 0.0372.

Standard CBM error_increase: saliency mean -0.0698; saliency-random mean 0.1330; 95% range [-2.5445, 2.3398]; saliency > random mean rate 0.5447.

Mixed-type CEM output_sensitivity: saliency mean 0.0569; saliency-random mean -0.3735; 95% range [-1.5450, 0.0348]; saliency > random mean rate 0.0579.

Mixed-type CEM error_increase: saliency mean -0.0285; saliency-random mean -0.0372; 95% range [-1.2431, 0.8668]; saliency > random mean rate 0.4874.

Learned-softmax GAM output_sensitivity: saliency mean 0.1919; saliency-random mean -1.1230; 95% range [-3.3577, 0.0803]; saliency > random mean rate 0.0447.

Learned-softmax GAM error_increase: saliency mean -0.1031; saliency-random mean 0.1941; 95% range [-2.3412, 2.6131]; saliency > random mean rate 0.5263.

Complete fold-target, pooled-target, and pooled-model results are retained in Table 8 (spatial_faithfulness.csv).

10. Discussion and Limitations

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

Learned-softmax GAM had the lowest pooled MAE point estimate, but not every paired CI excluded 0.

Intervention benefit was model-dependent; positive improvement was not uniform across $k=0-8$.

Saliency was not uniformly more faithful than matched random masks for either faithfulness quantity.

Unclipped Black-box scores extended below 1 and above 5; Standard CBM and Mixed-type CEM also produced a small fraction below 1, whereas Learned-softmax GAM stayed within the rating range.

All 6,955 undefined maps were confirmed post-ReLU zeros, but their full pre-ReLU/ gradient decomposition was not persisted.

The concentration of zero maps is classified SYSTEMATIC_MODEL/ TARGET_ISSUE rather than a proven implementation bug.

11. Reproducibility

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

All public values, CIs, tables, and figures derive from 1 shared report_data.json model.

GitHub excludes checkpoints, private predictions, raw Grad-CAM maps, CT/ ROI volumes, UIDs, and patient keys.

The private archive is verified file-by-file with SHA-256 and is stored only on the Mac.

Verified private archive: 1698 files; 14386651621 bytes; manifest

67731d14d26d5ff1cbbf36afa903490662f7c130abbc76421a3ebd39edf37df4.

Table 1: primary_secondary_metrics.csv.

Table 2: paired_comparisons.csv.

Table 3: concept_metrics.csv.

Table 4: intervention_curves.csv.

Table 5: centered_contributions.csv.

Table 6: learned_gam_alpha.csv.

Table 7: gradcam_accounting.csv.

Table 8: spatial_faithfulness.csv.

Table 9: execution_registry.csv.

12. References

This page summarizes frozen, deidentified P5–P9 evidence. No new model forward or scientific job was used.

- [1] S. G. Armato III et al., "The Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI): A completed reference database of lung nodules on CT scans," *Med. Phys.*, vol. 38, no. 2, pp. 915-931, 2011, doi: 10.1118/1.3528204.
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 - [6] B. Efron, "Bootstrap Methods: Another Look at the Jackknife," *Ann. Stat.*, vol. 7, no. 1, pp. 1-26, 1979, doi: 10.1214/aos/1176344552.
- Scientific conclusion code: GAM_LOWEST_POINT_ESTIMATE_MAE.
- Scientific conclusion code: PAIRED_MAE_SUPPORTS_GAM_OVER_BLACKBOX_AND_CBM.
- Scientific conclusion code: AUROC_DIFFERENCES_MOSTLY_UNCERTAIN.
- Scientific conclusion code: INTERVENTION_BENEFIT_MODEL_DEPENDENT.
- Scientific conclusion code: SALIENCY_NOT_UNIFORMLY_MORE_FAITHFUL_THAN_RANDOM.
- Scientific conclusion code: SYSTEMATIC_MODEL_TARGET_ZERO_MAP_LIMITATION.